

MS4215: Lab 1 Solutions

Spotify Track Energy

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Introduction

The dataset `lab1_dataset_spotify.csv` contains information on 114,000 tracks from Spotify, covering a wide range of genres. The variables include:

Variable	Description
<code>track_genre</code>	Genre of the track
<code>energy</code>	Intensity and activity of the track (0–1)
<code>loudness</code>	Overall loudness in decibels (dB)
<code>acousticness</code>	Likelihood the track is acoustic (0–1)
<code>valence</code>	Musical positiveness of the track (0–1)
<code>danceability</code>	How suitable the track is for dancing (0–1)
<code>tempo</code>	Estimated tempo in beats per minute (BPM)
<code>speechiness</code>	Presence of spoken words in the track (0–1)
<code>popularity</code>	Track popularity score (0–100)
<code>explicit</code>	Whether the track has explicit lyrics (TRUE/FALSE)
<code>duration_ms</code>	Duration of the track in milliseconds

The outcome variable for this lab is **energy**, which measures the intensity and activity of a track. A high energy value (close to 1) indicates a fast, loud, and noisy track; a low value (close to 0) indicates a slower, quieter track.

Before attempting the questions, run the setup chunk below. This loads the required packages and draws a random sample of 3,000 tracks for analysis.

```
#install.packages("tidyverse")
#install.packages("corrplot")
library(tidyverse)
library(corrplot)

set.seed(42)

# In the line of code below, you will need to change the location to where you
# have saved the file
spotify_full <- read_csv("data/lab1_dataset_spotify.csv")
spotify      <- spotify_full |> sample_n(3000)
```

Questions

(i) Explore the data

```
str(spotify)
```

```
## spc_tbl_ [3,000 x 20] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
## $ track_id      : chr [1:3000] "0elTi2owjDA2QWAFywLvff" "7ri2wXigElmFVgF5CAcrSF" "4SDLH34rwro1NgA" ...
## $ artists       : chr [1:3000] "AKB48" "Rival Consoles" "Osho Jain" "Luiz Gonzaga" ...
## $ album_name    : chr [1:3000] "GIVE ME FIVE! Type-A( )" "LUZoSCURA" "Kya Pata" "A Triste Partida" ...
## $ track_name    : chr [1:3000] "NEW SHIP( : A)" "Not Really - Mixed" "Kya Pata" "Numa Sala de" ...
## $ popularity    : num [1:3000] 23 11 37 44 22 0 16 21 45 43 ...
## $ duration_ms   : num [1:3000] 289560 201533 207272 138640 211746 ...
## $ explicit      : logi [1:3000] FALSE FALSE FALSE FALSE FALSE FALSE ...
## $ danceability  : num [1:3000] 0.374 0.653 0.756 0.583 0.586 0.657 0.56 0.666 0.844 0.713 ...
## $ energy        : num [1:3000] 0.911 0.97 0.308 0.513 0.758 0.817 0.247 0.471 0.52 0.236 ...
## $ key           : num [1:3000] 10 4 7 11 2 0 2 3 7 1 ...
## $ loudness      : num [1:3000] -2.74 -7.88 -11.35 -8.62 -12.49 ...
## $ mode          : num [1:3000] 1 0 1 0 1 1 1 1 1 1 ...
## $ speechiness   : num [1:3000] 0.0438 0.0427 0.0461 0.047 0.0472 0.0554 0.0601 0.0301 0.0518 0.03 ...
## $ acousticness  : num [1:3000] 0.00586 0.00316 0.809 0.637 0.00124 0.262 0.924 0.316 0.00139 0.88 ...
## $ instrumentalness: num [1:3000] 1.56e-03 8.90e-01 0.00 0.00 2.49e-01 6.79e-03 9.57e-01 5.54e-06 7. ...
## $ liveness      : num [1:3000] 0.324 0.412 0.346 0.352 0.0633 0.248 0.0918 0.113 0.0941 0.091 ...
## $ valence       : num [1:3000] 0.411 0.27 0.11 0.955 0.49 0.651 0.69 0.662 0.197 0.647 ...
## $ tempo         : num [1:3000] 166 124 132 152 166 ...
## $ time_signature : num [1:3000] 4 4 3 4 4 4 4 3 3 ...
## $ track_genre   : chr [1:3000] "j-idol" "idm" "singer-songwriter" "mpb" ...
## - attr(*, "spec")=
## .. cols(
## ..   track_id = col_character(),
## ..   artists = col_character(),
## ..   album_name = col_character(),
## ..   track_name = col_character(),
## ..   popularity = col_double(),
## ..   duration_ms = col_double(),
## ..   explicit = col_logical(),
## ..   danceability = col_double(),
## ..   energy = col_double(),
## ..   key = col_double(),
## ..   loudness = col_double(),
## ..   mode = col_double(),
## ..   speechiness = col_double(),
## ..   acousticness = col_double(),
## ..   instrumentalness = col_double(),
## ..   liveness = col_double(),
## ..   valence = col_double(),
## ..   tempo = col_double(),
## ..   time_signature = col_double(),
## ..   track_genre = col_character()
## .. )
## - attr(*, "problems")=<pointer: 0x860fec120>
```

```
summary(spotify)
```

```
##      track_id      artists      album_name      track_name
## Length :3000    Length :3000    Length :3000    Length :3000
```

```
## N.unique :2970 N.unique :2415 N.unique :2665 N.unique :2884
## N.blank : 0 N.blank : 0 N.blank : 0 N.blank : 0
## Min.nchar: 22 Min.nchar: 2 Min.nchar: 1 Min.nchar: 1
## Max.nchar: 22 Max.nchar: 173 Max.nchar: 124 Max.nchar: 174
##
## popularity duration_ms explicit danceability
## Min. : 0.00 Min. : 17826 Mode :logical Min. :0.0000
## 1st Qu.:16.00 1st Qu.: 174193 FALSE:2757 1st Qu.:0.4557
## Median :35.00 Median : 214573 TRUE :243 Median :0.5760
## Mean :32.83 Mean : 228215 Mean :0.5635
## 3rd Qu.:49.00 3rd Qu.: 261114 3rd Qu.:0.6900
## Max. :97.00 Max. :1800520 Max. :0.9760
##
## energy key loudness mode
## Min. :0.0000203 Min. : 0.00 Min. : -41.531 Min. :0.000
## 1st Qu.:0.4710000 1st Qu.: 2.00 1st Qu.: -9.950 1st Qu.:0.000
## Median :0.6850000 Median : 5.00 Median : -6.965 Median :1.000
## Mean :0.6433781 Mean : 5.34 Mean : -8.177 Mean :0.649
## 3rd Qu.:0.8590000 3rd Qu.: 9.00 3rd Qu.: -4.926 3rd Qu.:1.000
## Max. :0.9990000 Max. :11.00 Max. : 1.028 Max. :1.000
##
## speechiness acousticness instrumentalness liveness
## Min. :0.00000 Min. :1.130e-06 Min. :0.0000000 Min. :0.0176
## 1st Qu.:0.03580 1st Qu.:1.700e-02 1st Qu.:0.0000000 1st Qu.:0.0983
## Median :0.04795 Median :1.740e-01 Median :0.0000307 Median :0.1360
## Mean :0.08306 Mean :3.148e-01 Mean :0.1526615 Mean :0.2176
## 3rd Qu.:0.08052 3rd Qu.:5.990e-01 3rd Qu.:0.0301000 3rd Qu.:0.2840
## Max. :0.94300 Max. :9.960e-01 Max. :0.9930000 Max. :1.0000
##
## valence tempo time_signature track_genre
## Min. :0.0000 Min. : 0.00 Min. :0.000 Length :3000
## 1st Qu.:0.2580 1st Qu.: 99.93 1st Qu.:4.000 N.unique : 114
## Median :0.4640 Median :122.02 Median :4.000 N.blank : 0
## Mean :0.4750 Mean :122.64 Mean :3.894 Min.nchar: 3
## 3rd Qu.:0.6823 3rd Qu.:140.32 3rd Qu.:4.000 Max.nchar: 17
## Max. :0.9880 Max. :209.56 Max. :5.000
```

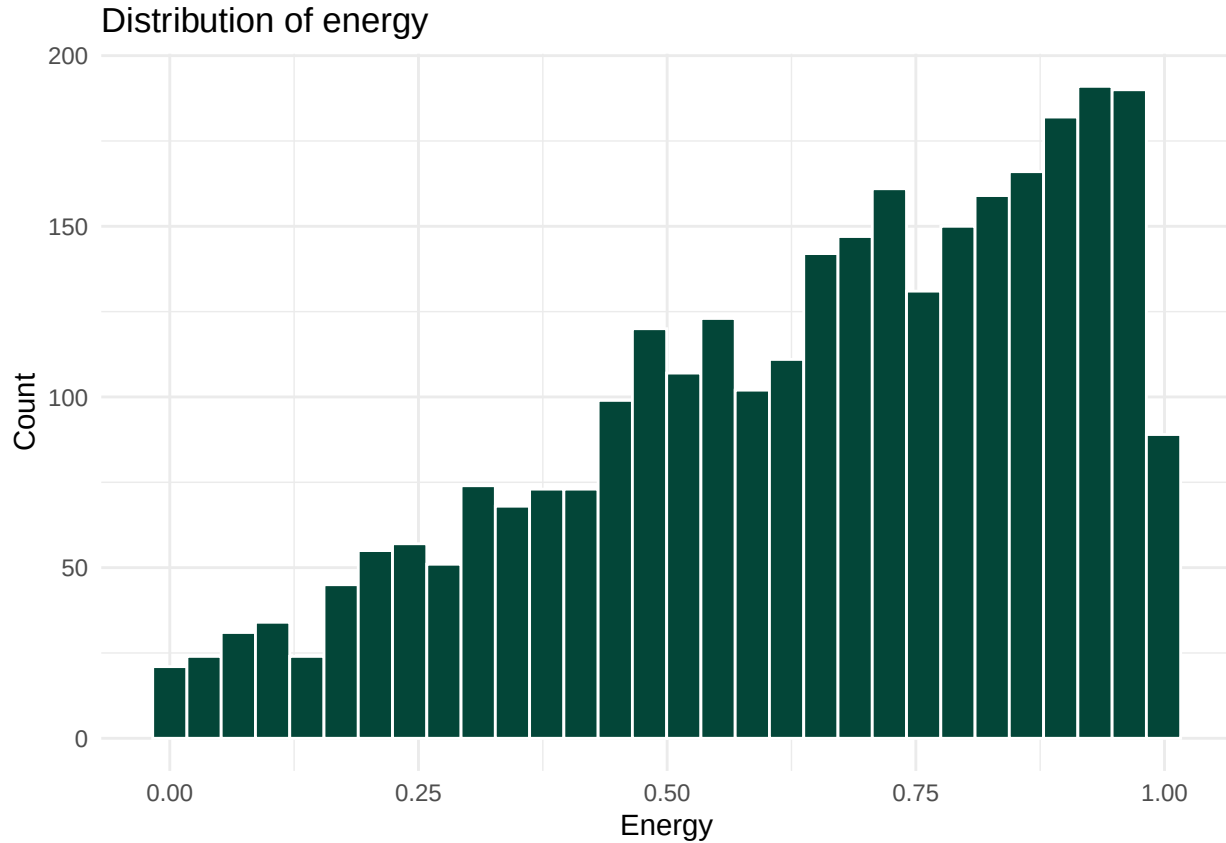
```
range(spotify$energy)
```

```
## [1] 2.03e-05 9.99e-01
```

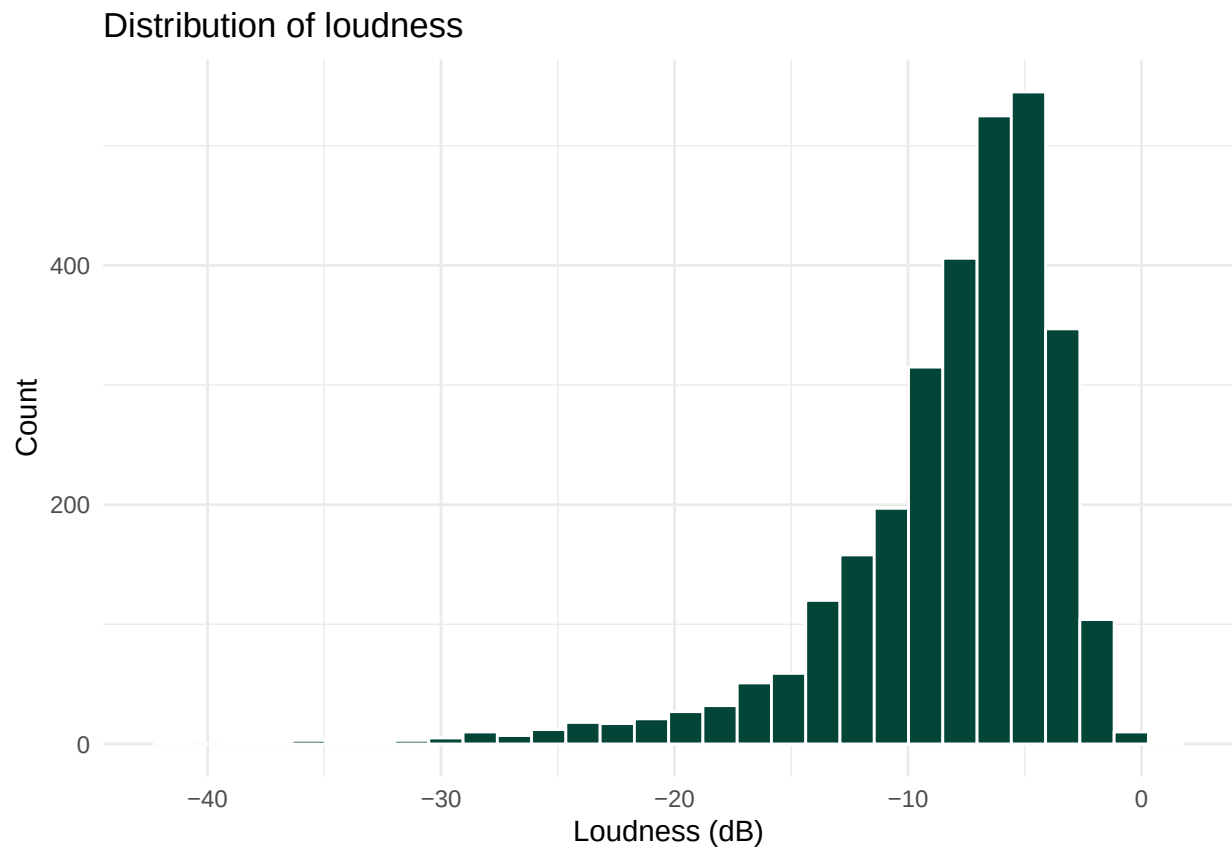
Energy ranges from 0 to 1 across the full scale. The `str()` output shows the variable types; `summary()` provides the five-number summary plus mean for numeric variables and counts for character variables.

(ii) Histograms and summary statistics

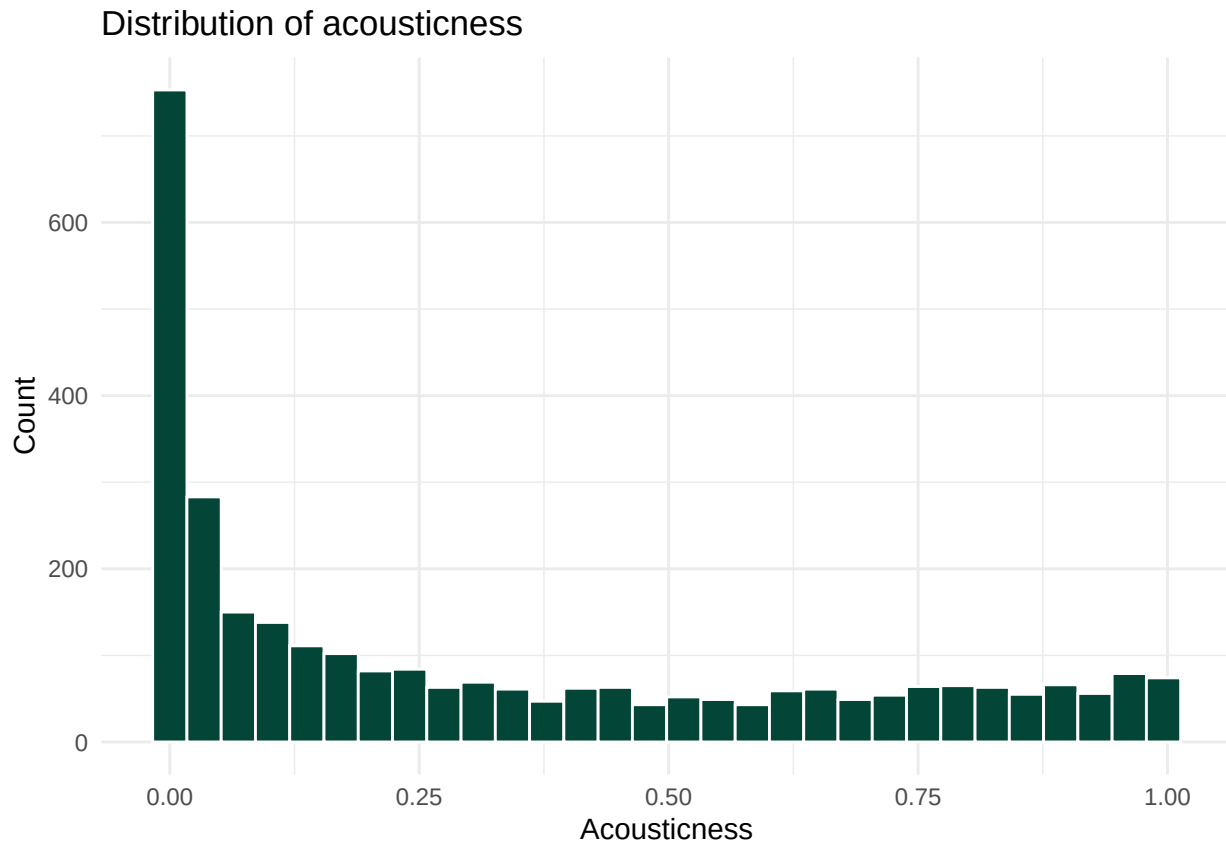
```
ggplot(spotify, aes(x = energy)) +  
  geom_histogram(bins = 30, fill = "#034638", colour = "white") +  
  labs(x = "Energy", y = "Count", title = "Distribution of energy") +  
  theme_minimal()
```



```
ggplot(spotify, aes(x = loudness)) +  
  geom_histogram(bins = 30, fill = "#034638", colour = "white") +  
  labs(x = "Loudness (dB)", y = "Count", title = "Distribution of loudness") +  
  theme_minimal()
```



```
ggplot(spotify, aes(x = acousticness)) +  
  geom_histogram(bins = 30, fill = "#034638", colour = "white") +  
  labs(x = "Acousticness", y = "Count", title = "Distribution of acousticness") +  
  theme_minimal()
```



```
spotify |> select(energy, loudness, acousticness) |> summary()
```

##	energy	loudness	acousticness
## Min.	:0.0000203	Min. : -41.531	Min. :1.130e-06
## 1st Qu.:	0.4710000	1st Qu.: -9.950	1st Qu.:1.700e-02
## Median :	0.6850000	Median : -6.965	Median :1.740e-01
## Mean :	0.6433781	Mean : -8.177	Mean :3.148e-01
## 3rd Qu.:	0.8590000	3rd Qu.: -4.926	3rd Qu.:5.990e-01
## Max. :	0.9990000	Max. : 1.028	Max. :9.960e-01

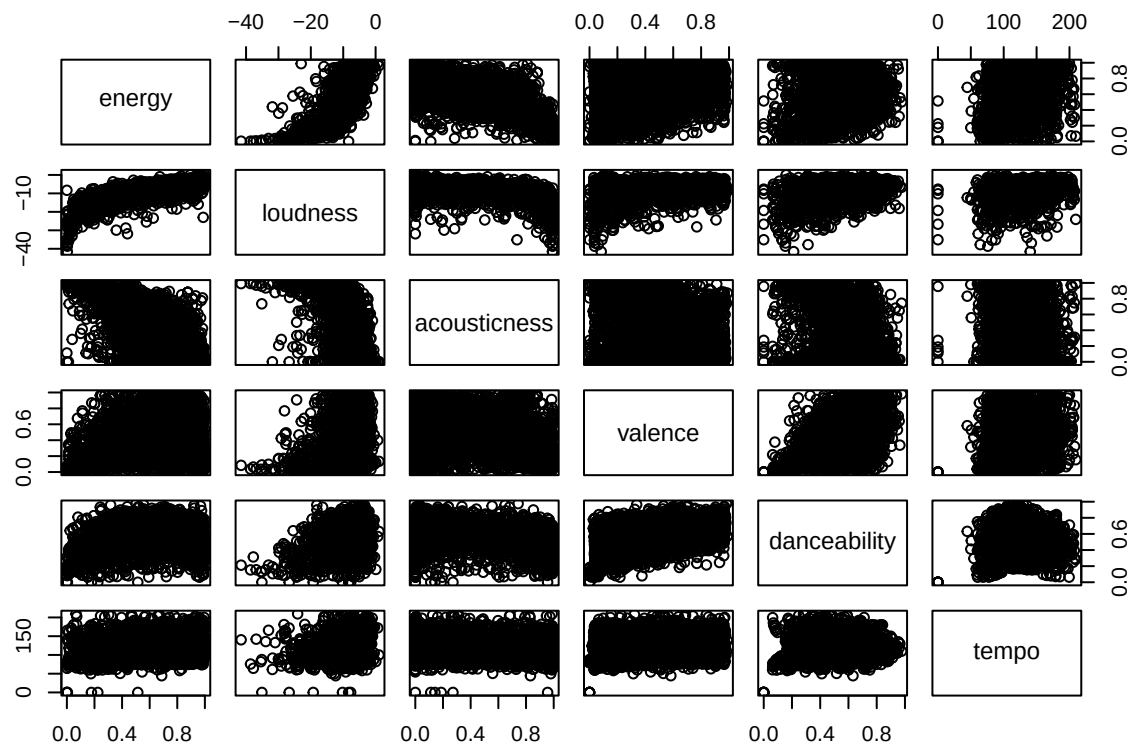
Energy: The distribution is moderately left-skewed; the median (0.685) is above the mean (0.643), with more tracks concentrated at higher energy levels and a longer left tail. Report **median and IQR**.

Loudness: Measured in decibels (dB, always ≤ 0). The distribution is left-skewed, with most tracks clustering between -12 dB and 0 dB and a long left tail of very quiet tracks. The median (-6.97 dB) is above the mean (-8.18 dB). Report **median and IQR**.

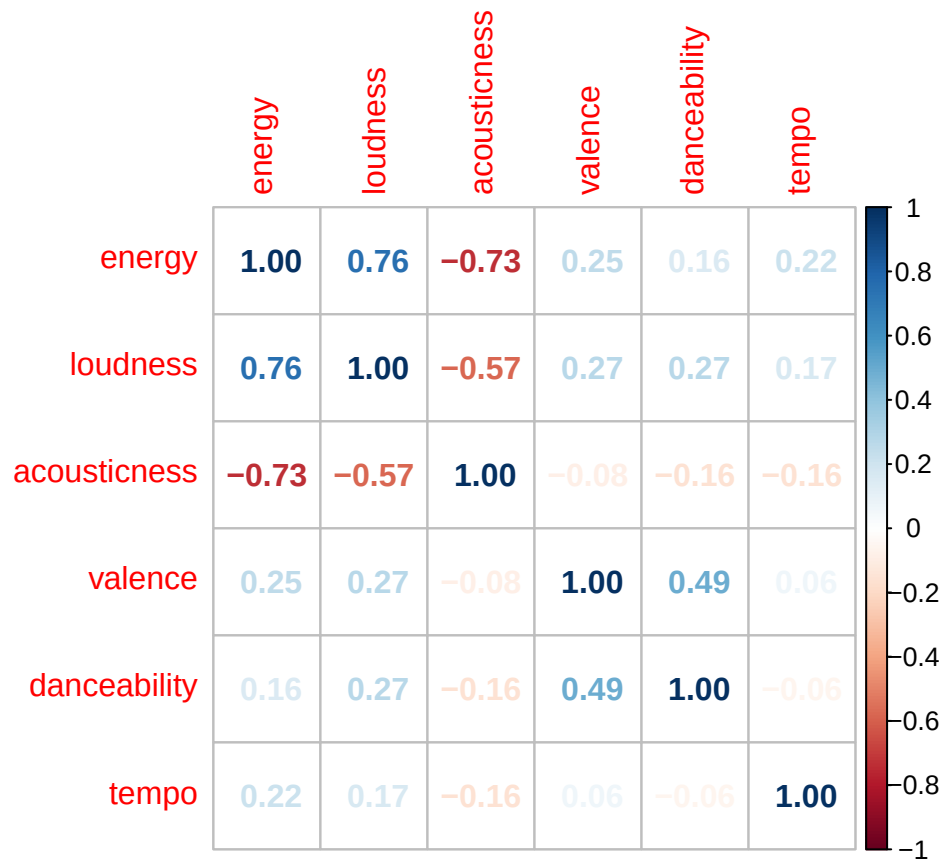
Acousticness: The distribution is strongly right-skewed; most tracks have acousticness close to 0 (not acoustic), with a long right tail of highly acoustic tracks. The mean (0.315) is well above the median (0.174). Report **median and IQR**.

(iii) Scatter plot matrix and correlation

```
vars <- c("energy", "loudness", "acousticness", "valence", "danceability", "tempo")  
pairs(spotify |> select(all_of(vars)))
```



```
R_mat <- spotify |> select(all_of(vars)) |> cor(use = "complete.obs")  
corrplot(R_mat, method = "number")
```



loudness has the strongest positive correlation with energy ($r = 0.76$). The relationship is positive: louder tracks tend to be more energetic. The correlation is strong. acousticness has the strongest negative correlation ($r = -0.73$): acoustic tracks tend to have low energy, which makes intuitive sense.

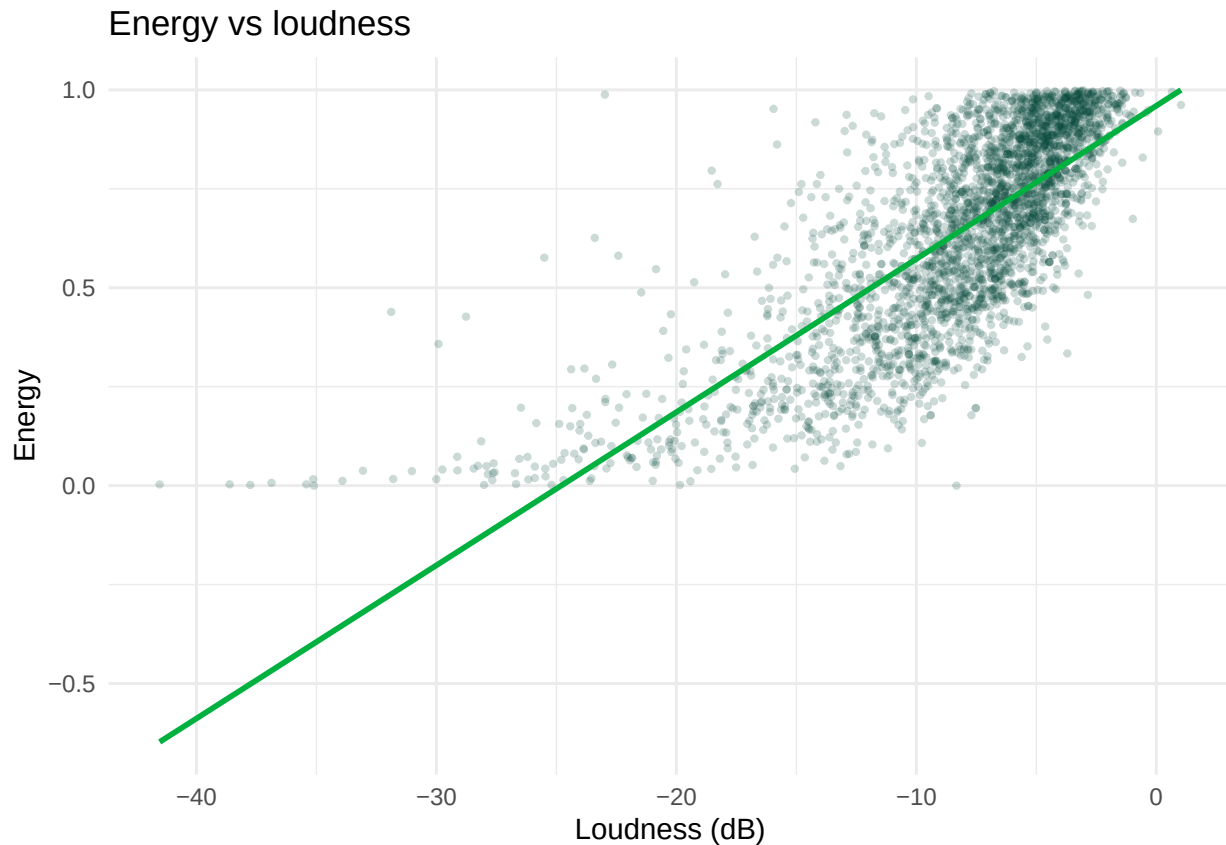
(iv) Simple linear regression

```
m1 <- lm(energy ~ loudness, data = spotify)
summary(m1)

##
## Call:
## lm(formula = energy ~ loudness, data = spotify)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.63754 -0.11433  0.01029  0.11322  0.91762
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.9598753  0.0057862  165.89  <2e-16 ***
## loudness     0.0387074  0.0006053   63.95  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1642 on 2998 degrees of freedom
## Multiple R-squared:  0.577, Adjusted R-squared:  0.5769
## F-statistic: 4090 on 1 and 2998 DF, p-value: < 2.2e-16

ggplot(spotify, aes(x = loudness, y = energy)) +
  geom_point(alpha = 0.2, colour = "#034638", size = 0.8) +
  geom_smooth(method = "lm", se = FALSE, colour = "#00B140") +
  labs(
    x = "Loudness (dB)",
    y = "Energy",
    title = "Energy vs loudness"
  ) +
  theme_minimal()

## `geom_smooth()` using formula = 'y ~ x'
```



Fitted model: $\widehat{\text{energy}} = 0.960 + 0.039 \times \text{loudness}$

- **Intercept (0.960):** the predicted energy when loudness = 0 dB (maximum possible loudness). Not practically meaningful since loudness is always negative, but it anchors the line.
- **Slope (0.039):** for each 1 dB increase in loudness (i.e., the track gets louder), predicted energy increases by 0.039 on average. Since loudness values are negative, a track at -5 dB is predicted to be more energetic than one at -10 dB.
- **$R^2 = 0.577$:** Loudness explains approximately 57.7% of the variation in energy, which is a strong fit for a single predictor.

(v) Multiple linear regression

```
m2 <- lm(energy ~ loudness + acousticness + valence + danceability, data = spotify)
summary(m2)
```

```
##
## Call:
## lm(formula = energy ~ loudness + acousticness + valence + danceability,
##     data = spotify)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.74378 -0.08878  0.00226  0.08583  0.64340
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.992317   0.010539   94.16  <2e-16 ***
## loudness       0.025160   0.000624   40.32  <2e-16 ***
## acousticness -0.348314   0.008995  -38.72  <2e-16 ***
## valence        0.133449   0.010848   12.30  <2e-16 ***
## danceability -0.172073   0.016228  -10.60  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1327 on 2995 degrees of freedom
## Multiple R-squared:  0.7241, Adjusted R-squared:  0.7237
## F-statistic: 1965 on 4 and 2995 DF,  p-value: < 2.2e-16
```

Interpretation of partial slopes:

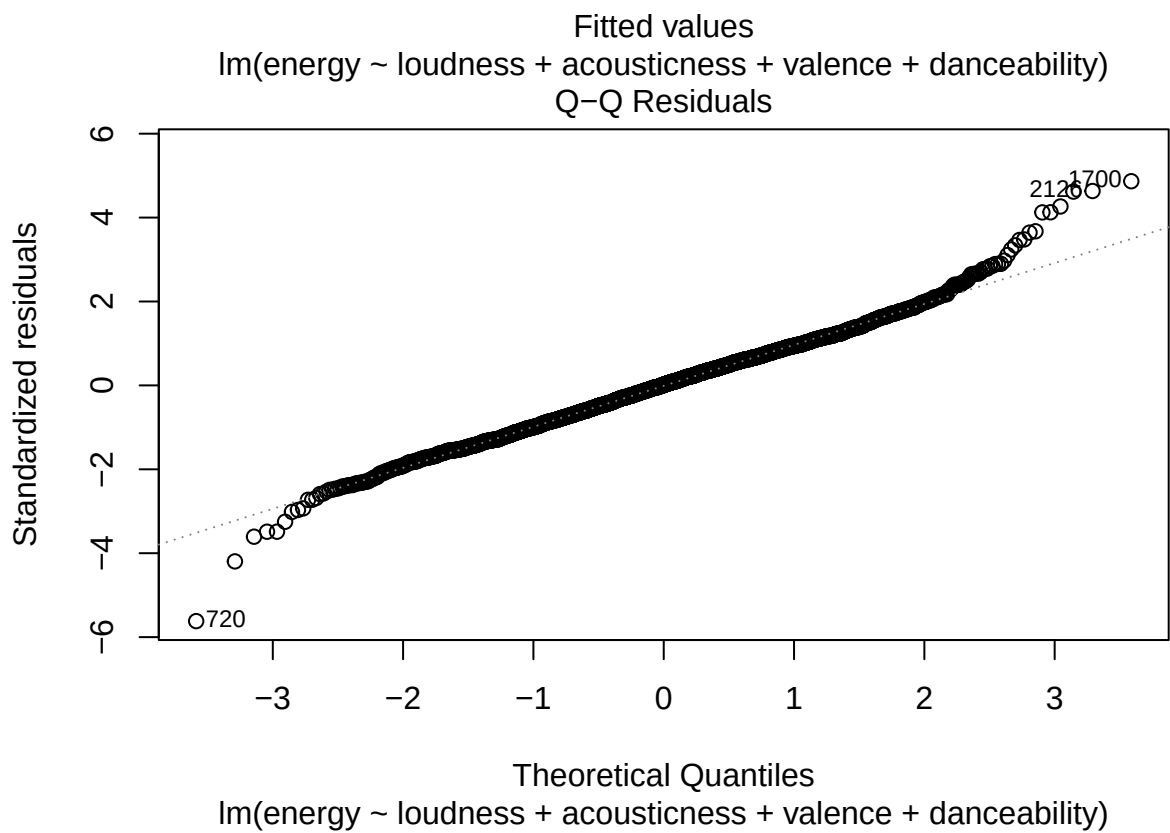
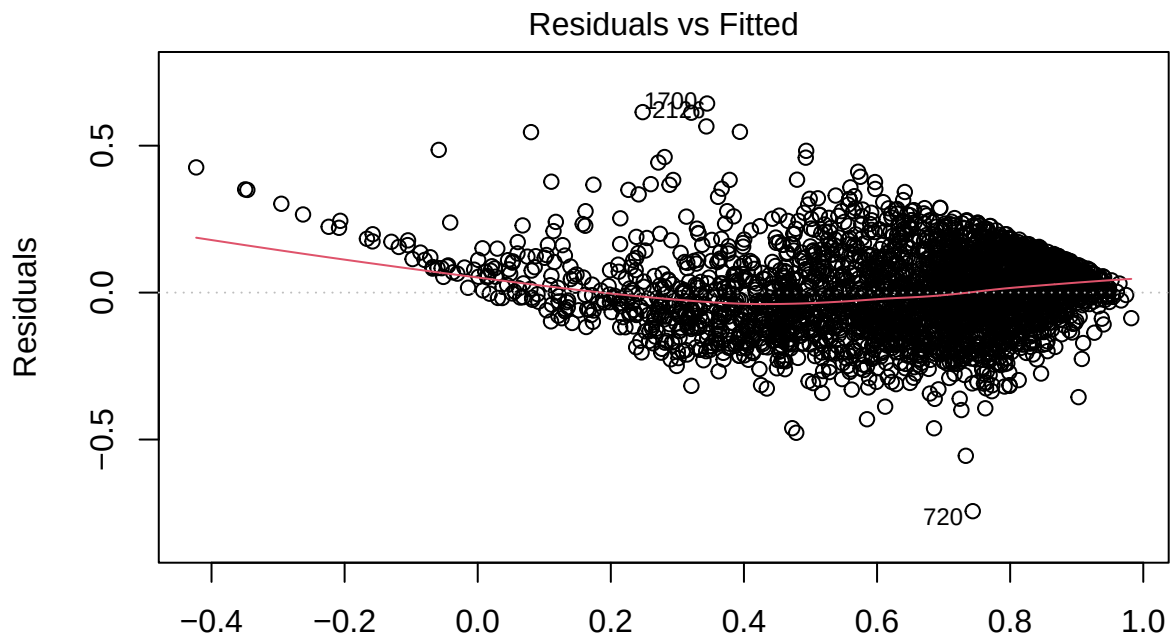
- **loudness (0.025):** holding acousticness, valence, and danceability constant, each 1 dB increase in loudness is associated with a 0.025 increase in energy on average.
- **acousticness (−0.348):** holding other predictors constant, a one-unit increase in acousticness (fully acoustic track vs fully non-acoustic) is associated with a 0.348 decrease in energy on average. Acoustic tracks are substantially less energetic.
- **valence (0.133):** holding other predictors constant, a one-unit increase in valence (more positive/happy) is associated with a 0.133 increase in energy on average.
- **danceability (−0.172):** holding other predictors constant, a one-unit increase in danceability is associated with a 0.172 decrease in energy on average. After accounting for loudness, acoustic, and emotional character, more danceable tracks tend to be slightly less intense.

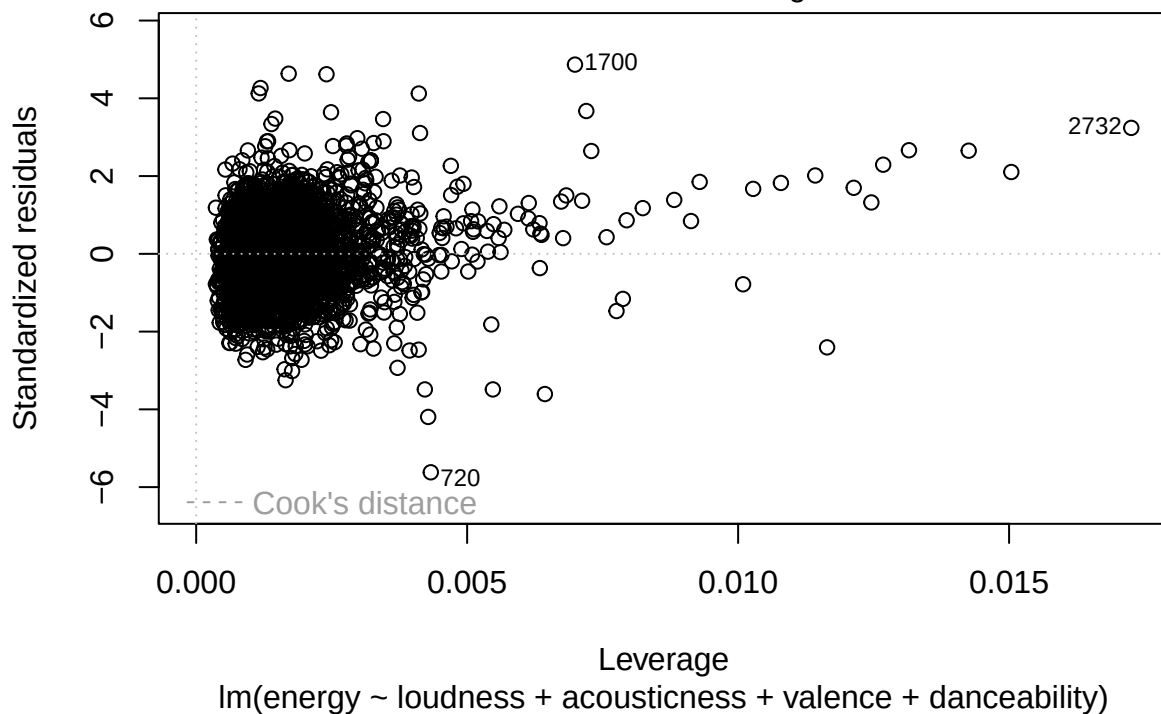
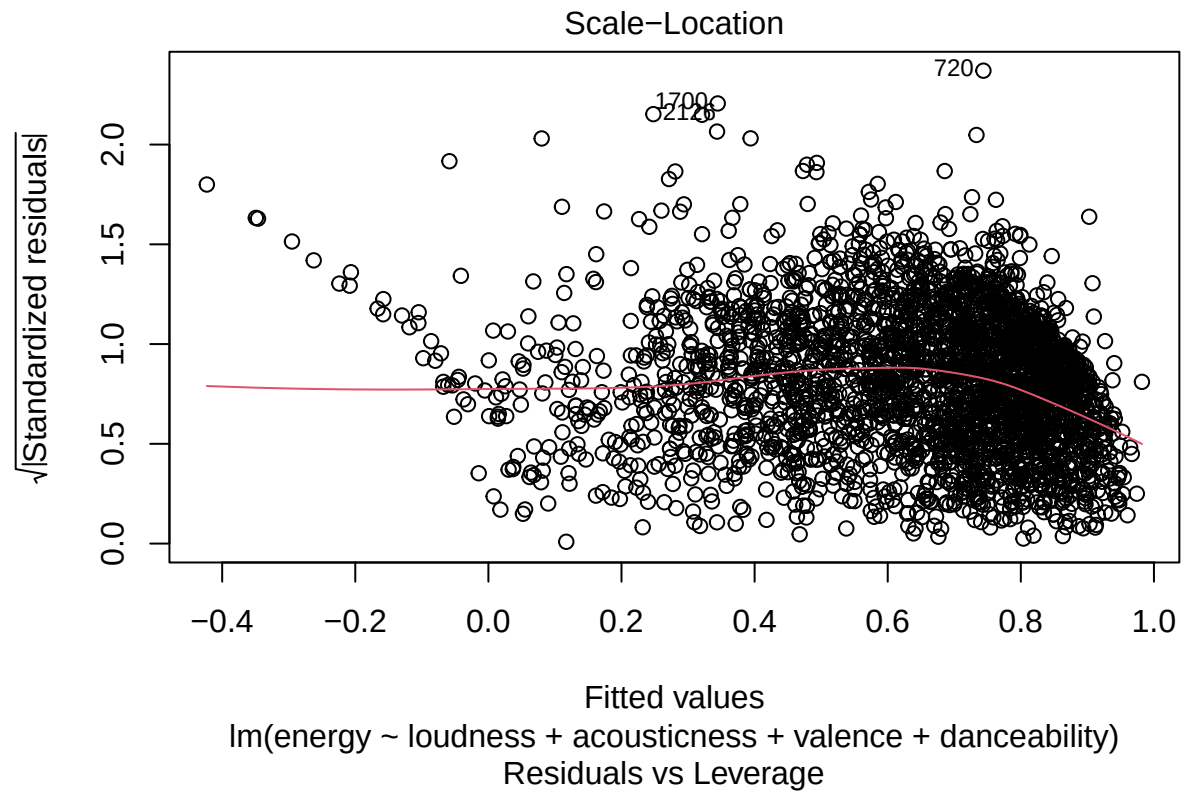
The adjusted R^2 is 0.724, which is higher than the SLR in (iv) (0.577), indicating that acousticness, valence, and danceability each contribute additional explanatory power beyond loudness alone.

All four predictors have $p < 0.001$ and are statistically significant at the 5% level.

(vi) Residual diagnostics

```
plot(m2)
```





Residuals vs Fitted: Check for non-linearity, points should be randomly scattered around zero with no systematic pattern.

Normal Q-Q: Check for normality of residuals, points should follow the diagonal line closely.

Scale-Location: Check for homoscedasticity, the spread of residuals should be roughly constant across fitted values.

Residuals vs Leverage: Identify influential observations, points beyond Cook's distance lines warrant investigation.